

3. Algorithms on graphs II		
3.1	Algorithm for finding the shortest route around a network, travelling along every edge at least once and ending at the start vertex. The network will have up to four odd nodes.	Also known as the ‘Chinese postman’ problem. Students will be expected to use inspection to consider all possible pairings of odd nodes. (The application of Floyd’s algorithm to the odd nodes is not required.)
3.2	The practical and classical Travelling Salesman problems. The classical problem for complete graphs satisfying the triangle inequality.	The use of short cuts to improve upper bound is included.
3.3	Determination of upper and lower bounds using minimum spanning tree methods.	The conversion of a network into a complete network of shortest ‘distances’ is included.
3.4	The nearest neighbour algorithm.	